MIRROR IMAGES

Left-Right Reversal, Letters, Numbers and Figures

OAVS TGT | Reasoning (Non-Verbal) | Topic 16: seventh non-verbal topic

Syllabus line (Reasoning, 30 Marks): “The Non-Verbal Reasoning topics like Figure Formation, Dice, Triangle, Rule Detection, Images, Completion of Pattern, **Mirror Images**, Figure Matrix and Paper Folding, etc.”

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). **All 8 mirror-image PYQs are solved below with their original figures.** The practice set was generated by program, so each question has exactly one option that fits.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	All 8 PYQs, what each tests, priority list
2	Foundation	Core rules, symmetric letters and digits, figure changes, the 4-step method
3	Solved PYQs	8 PYQs with original figures and step-by-step reasoning
4	Shortcuts, Traps and Scope	Speed methods, common mistakes, what is not covered
5	Practice Set	8 original questions with a verified key
6	Quick Revision	One-page cheat sheet

CHAPTER 1

PYQ Analysis

8 questions in 5 papers: a guaranteed topic

Paper	Q.	Item mirrored	Mirror	Ans
English 2019	34	Scissors (curved, shaded figure)	Right (MN)	3
English 2019	46	Word CUP	Right (MN)	4
English 2022	35	Word BIOLOGY	Right	A
Mathematics 2024	77	Scallops + line + arrow + arrowhead	Right (MN)	C
Mathematics 2024	79	Boxes with diagonal and black rectangle	Right (MN)	A
Mathematics 2024	93	Triangles, slanted lines, ladder	Right (MN)	A
Science 2024	97	Alphanumeric 47hREAZ	Right (MN)	1
Social Studies 2023	4	Letters gfQNham	Right (PQ)	2

PYQ INSIGHT: Every PYQ so far uses a **vertical mirror on the right** (left-right reversal). Four are words/alphanumerics and four are line figures (arrows, shading, slanted lines). Mathematics 2024 alone had three, so expect 1 to 3 questions in every paper. The wrong options are always made by **one of four faults**: unchanged figure, 180° turn, up-down flip, or flipped parts in the wrong places.

Pattern	PYQs	Priority
Word / alphanumeric, mirror at right	4	★★★ Must
Line figure with arrows, shading, slanted lines, mirror at right	4	★★★ Must
Mirror placed above or below (horizontal mirror)	0	★★ Expected
Mirror placed at the left	0	★★ Expected
Word that looks the same in its image (symmetric letters)	0	★ Possible

CHAPTER 2

Foundation

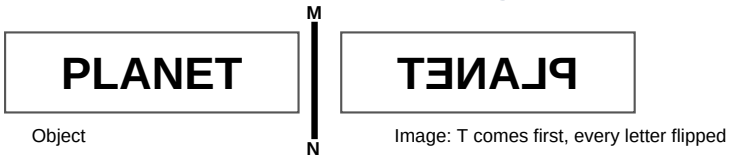
Rules, symmetric letters and the 4-step method

2.1 Core idea

A plane mirror reverses the side that faces it. The part of the object **nearest the mirror stays nearest** in the image, and the image is the same distance behind the mirror. The mirror image is **not** a rotation of the object.

Mirror position	What swaps	What does NOT change	Words
At the right or left (vertical mirror) – the usual PYQ case	Left and right	Top and bottom, heights, shading	Order of letters reverses; each letter flips left-right
Above or below (horizontal mirror)	Top and bottom	Left and right	Order stays; each letter turns upside-down

2.2 Worked picture (mirror at right)



2.3 Symmetric characters (they look the same after the flip)

Mirror	Letters unchanged	Digits unchanged
Vertical (side mirror)	A H I M O T U V W X Y	0, 1, 8
Horizontal (top/bottom mirror)	B C D E H I K O X	0, 1, 3, 8
Both	H I O X	0, 1, 8

NOTE: Digits 2, 3, 4, 5, 6, 7, 9 and letters such as R, E, F, G, J, K, L, N, P, Q, S, Z change in a side mirror. The mirror image of **6 is a reversed 6, not 9** (9 comes from a 180° turn).

2.4 Figures: what changes in a side mirror

Part of the figure	In the image
Position left / right	Swapped (the part nearest the mirror stays nearest)
Position up / down	Same
Arrow pointing left or right	Reverses (→ becomes ←)
Arrow pointing up or down	Same
Diagonal “\” or “/”	Slant reverses (\ becomes /)
Clockwise curve or arrow	Becomes anticlockwise
Shading, black or white, inside or outside	Same
Shape, size, number of parts	Same

2.5 The 4-step method

Step	Do this
1	Note where the mirror is (right/left or above/below) so you know the axis.
2	Words: write the word backwards, then flip each asymmetric letter. Figures: swap the parts across the axis.
3	Check the part **nearest the mirror**: it must be the first thing in the image.
4	Eliminate options that are unchanged, turned 180° or up-down flipped, then check small details (arrow direction, slant, shading).

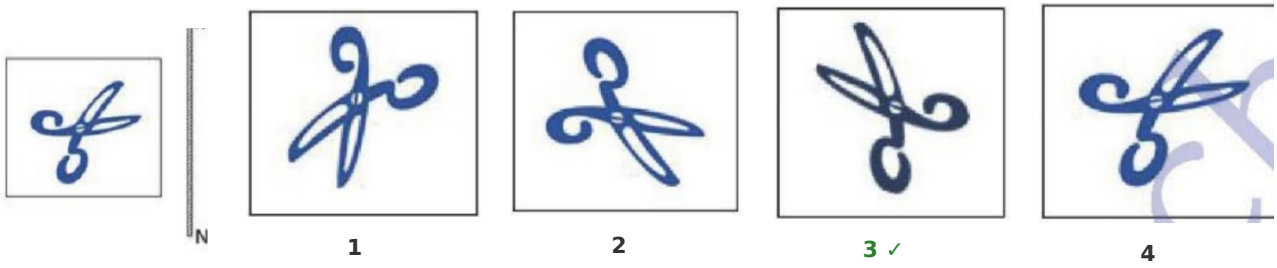
CHAPTER 3

Solved PYQs

All 8 questions with original figures

English 2019 Q34

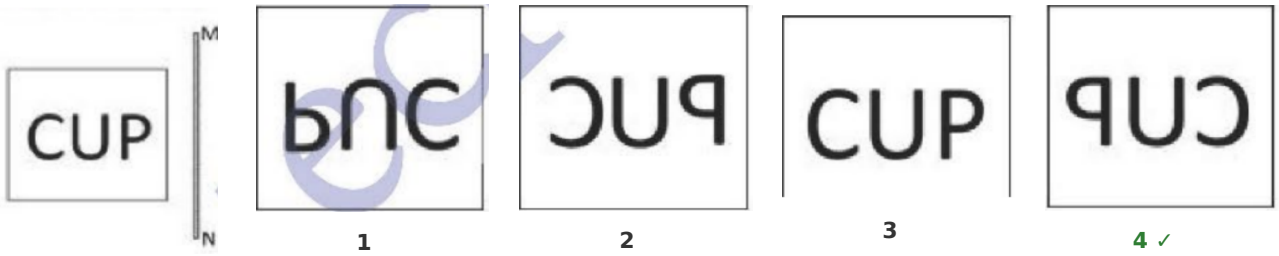
Identify the mirror image of the figure (mirror MN at the right).



1. Mirror is at the right, so left and right swap and up/down stays.
2. In the object the curled handles are at the **lower left** and the blades at the **upper right**.
3. In the image the handles must be at the **lower right** and the blades at the **upper left**.
4. Option 4 is the object itself (handles still on the left). Options 1 and 2 are turned sideways. Only option 3 has handles lower right, blades upper left.
Answer: Option 3

English 2019 Q46

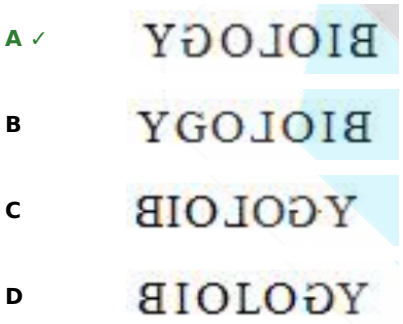
Identify the mirror image of CUP (mirror MN at the right).



1. P is the letter nearest the mirror, so the image starts with P and ends with C: order reverses.
2. P, U and C must each be flipped left-right (the bowl of P faces left, the opening of C faces right).
3. Option 3 is the unchanged word (rejected). Options 1 and 2 turn letters upside-down or place them wrongly. Option 4 is the only reversed, flipped word.
Answer: Option 4

English 2022 Q35

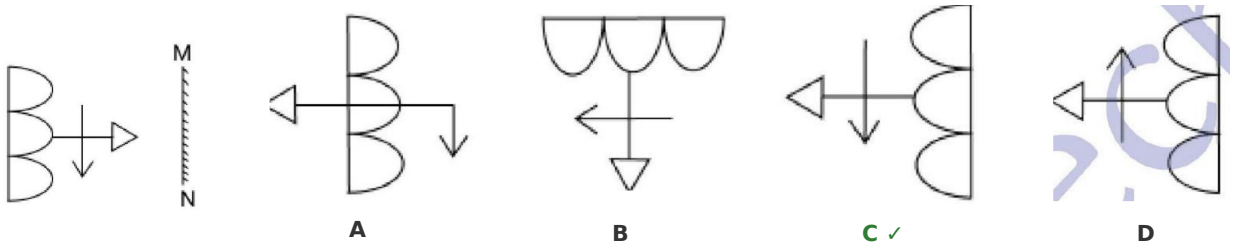
Select the correct mirror image of the word **BIOLOGY**. The mirror is placed to the right.



1. Right mirror: write BIOLOGY backwards: Y G O L O I B.
2. Flip the asymmetric letters B, L and G left-right; I, O and Y are symmetric and look the same. The image reads Y G O L O I B with B, L and G reversed.
3. Options B, C and D each differ from A in the orientation of at least one letter (left unflipped or turned upside-down), so only A passes the letter-by-letter check.
Answer: Option A

Mathematics 2024 Q77

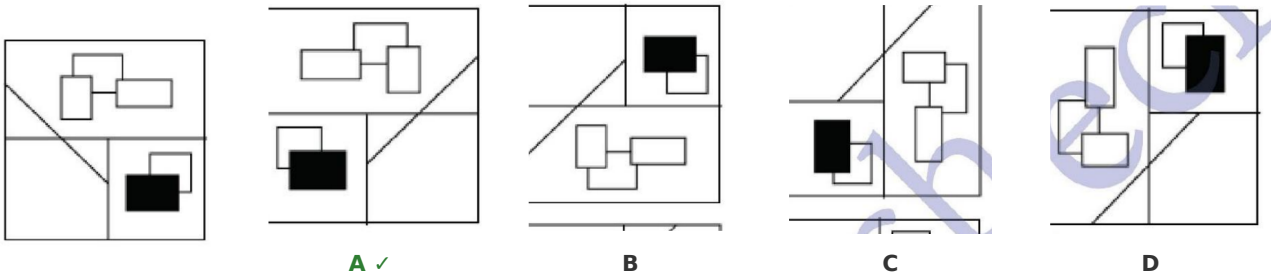
Select the mirror image of the figure (mirror MN at the right).



1. The scallops are at the left (farthest from the mirror); the line, down-arrow and triangle head are to the right (nearest).
2. Image: the triangle head goes to the far **left** and points left; the scallops go to the far **right** with their bulges facing left.
3. The down-arrow stays pointing **down** (up/down never changes in a side mirror).
4. D shows the arrow pointing up (wrong); B is turned sideways; A has the scallop stack in the middle with the arrow at the far right (parts not swapped across the axis). C matches.
Answer: Option C

Mathematics 2024 Q79

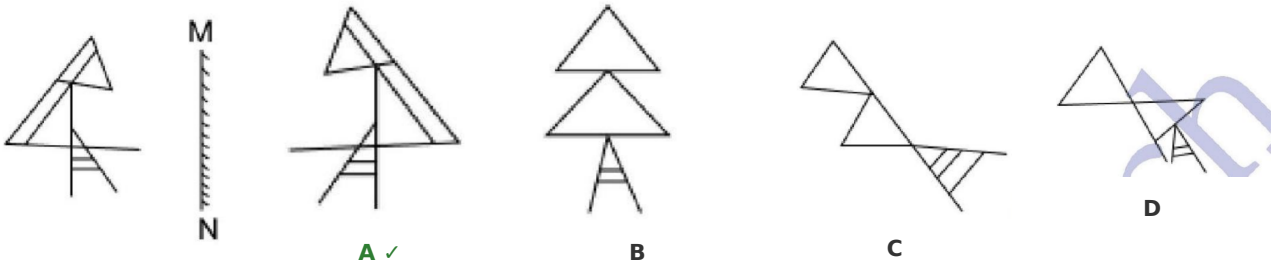
Select the mirror image of the figure (mirror MN at the right).



1. Object: connected white rectangles on top; a diagonal "\" crossing the middle line on the left; a black rectangle at the **bottom right**.
2. Image: the diagonal becomes "/" and moves to the right; the black rectangle moves to the **bottom left**; the white rectangles stay on top.
3. B and D put the black rectangle at the top (an up-down flip, wrong for a side mirror). C is turned sideways. Option A is correct.
Answer: Option A

Mathematics 2024 Q93

Select the mirror image of the figure (mirror MN at the right).



1. Object: a vertical stem with a ladder on its right at the bottom, a long horizontal line crossing the stem, and a triangle with slanted parallel lines at the top.
2. Image: the ladder moves to the **left** of the stem, the horizontal line extends to the other side, and every slanted line reverses its slant. The stem stays upright.
3. B is a different, symmetric figure; C and D are turned (not upright). Only A keeps the stem upright with the changes above.
Answer: Option A

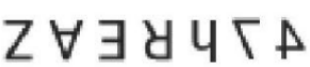
Science 2024 Q97

Select the mirror image of **47hREAZ** when the mirror MN is at the right.



1 ✓ 

2 

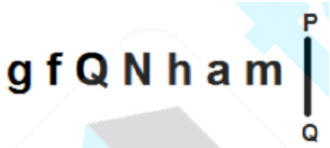
3 

4 

1. Reverse the order: Z, A, E, R, h, 7, 4 (Z is nearest the mirror, so it comes first).
2. Flip every asymmetric character: E, R, h, 7 and 4 are reversed left-right; A and Z keep their look (A is symmetric).
3. Option 2 has the order wrong, option 4 leaves the '4' unflipped and option 3 shows upside-down letters. Option 1 is correct.
Answer: Option 1

Social Studies 2023 Q4

Select the mirror image of **gfQNham** when the mirror PQ is at the right.



1 

2 ✓ 

3 

4 

1. Reverse the order: m, a, h, N, Q, f, g.
2. Flip each letter left-right while the heights stay: g and f keep their tops and tails but face the other way; N and Q reverse.
3. Options 1, 3 and 4 alter one or more letter shapes (an upside-down or wrongly flipped letter). Option 2 keeps heights and flips every letter. Option 2 is correct.
Answer: Option 2

> **SHORTCUT:** For every PYQ here the winning method was the same: **(1) reverse the order, (2) flip each asymmetric part, (3) check the part nearest the mirror**. Do not compare whole options at once; check the first and last items of each option.

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

• **Nearest-first rule:** the last letter of the word (or the part touching the mirror side) is the first in the image. This alone kills half the options.

• **Symmetric-letter shortcut:** in a word of symmetric letters (A H I M O T U V W X Y) only the order reverses, so look for the reversed word.

• **Two-point test for figures:** pick two distinctive parts (for example an arrowhead and a black shape). Their left-right order must swap while their heights stay.

• **Arrow direction:** left/right arrows reverse in a side mirror; up/down arrows do not.

• **Slant check:** every "\" becomes "/" in a side mirror; in a top/bottom mirror it also becomes "/".

• **Time:** 30 seconds per question is enough.

Trap	How to avoid it
Choosing the unchanged figure	Always look for a change; a mirror image is never identical unless the object is symmetric.
Choosing the 180° turn	Turning the page upside-down changes up and down too; a side mirror must not.
Reversing the order but not flipping letters	Each letter must also be flipped (CUP becomes P U C with each letter reversed).
Flipping letters but not reversing the order	The order always reverses in a side mirror.
Treating 6 as 9	A mirror image of 6 is a reversed 6; 9 comes from rotation.
Using the wrong axis	Check whether MN is vertical (side mirror) or horizontal (top/bottom mirror) first.
Forgetting that up/down arrows stay	Only left/right swap in a side mirror.

Scope

Related PYQs / topics	Where
"How would the pattern appear when the transparent sheet is folded at the dotted line?" (Eng 2019 Q47, Q49)	Paper Folding (next, with Figure Matrix)
Water images and clock mirrors	No PYQ in these five papers, so not covered here

CHAPTER 5

Practice Set

8 original questions; answers on the next page (every option set verified by program)

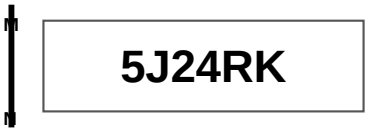
A. Words and alphanumerics (vertical mirror)

Q1. Select the mirror image of the word when the mirror MN is placed at the right.



A	B	C	D
TENALP	ꟻLANƎT	TƎNALꟻ	⊥ƎNAꟻ

Q2. Select the mirror image when the mirror MN is placed at the left.



A	B	C	D
ꓘꓤ4ƧſƧ	ƧſƧ4ꓤꓘ	KR42J5	ƧſƧ4ꓤK

B. Horizontal mirror and letter symmetry

Q3. Select the mirror image when the mirror MN is placed below the word.



A	B	C	D
TAOLꟻ	FLOAT	ꟻLOAT	⊥AOꟻ

Q4. Which word looks exactly the same in its mirror image when the mirror is placed at the right?

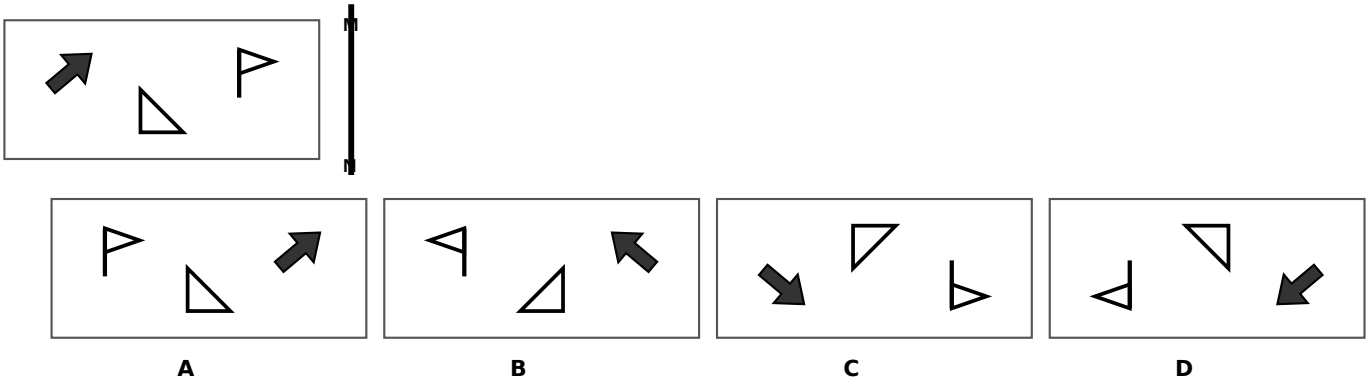
A. MAD B. MOM C. TOP D. HAM

Q5. Which word looks exactly the same in its mirror image when the mirror is placed above or below it?

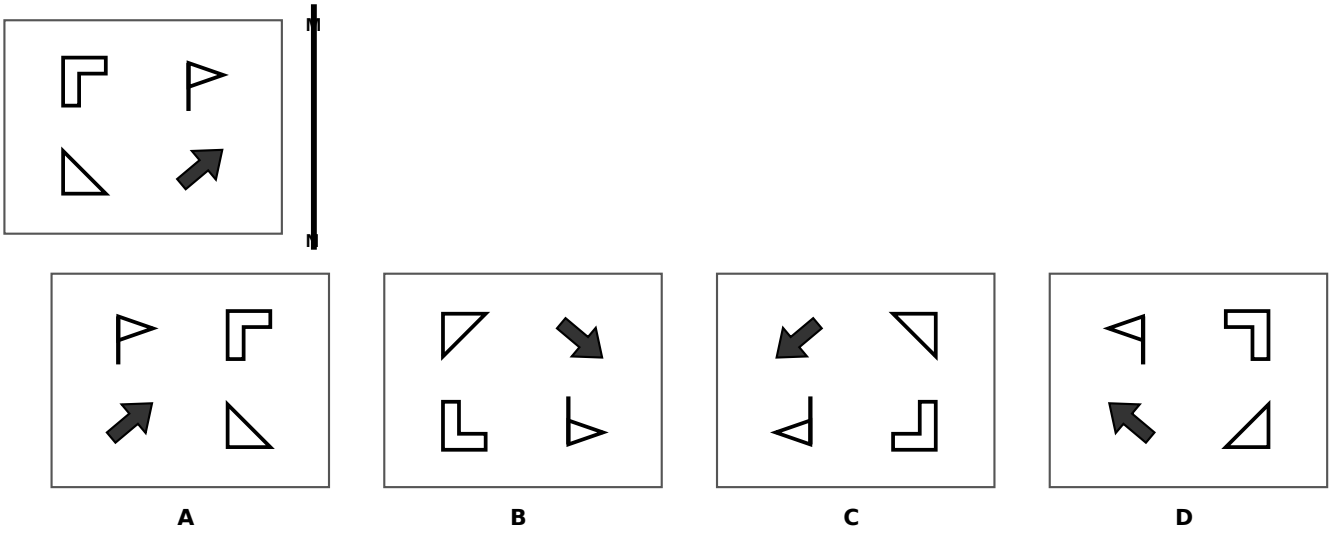
A. TAKE B. HOME C. BOOK D. COLD

C. Figures (position, direction and shape)

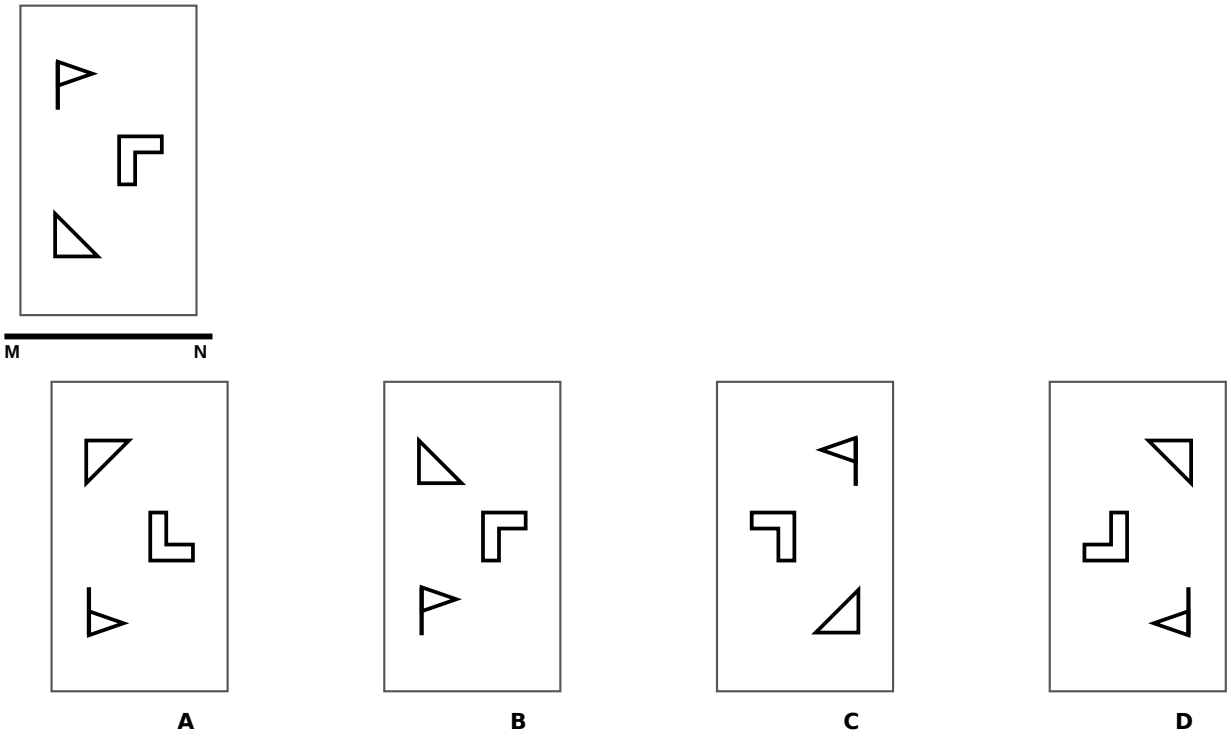
Q6. Select the mirror image of the figure when the mirror MN is placed at the right.



Q7. Select the mirror image of the figure when the mirror MN is placed at the right.



Q8. Select the mirror image of the figure when the mirror MN is placed <u>below</u> it.



Answer key

Q	Answer	Reason
1	C	Mirror at right: reverse the order (T first) and flip every letter. A has upright letters in reversed order; B has flipped letters in the old order; D is the word turned through 180°.
2	A	Mirror at the left works the same as at the right: left and right swap, so K comes first and each symbol is flipped. B is the up-down image (wrong mirror direction); C keeps symbols upright; D keeps the old order.
3	D	Mirror below: top and bottom swap, order of letters stays F-L-O-A-T and each letter turns upside-down. A is the side-mirror image, B is the unchanged word, C is the 180° turn.
4	B	In a side mirror a word looks unchanged only if it is a palindrome built from letters symmetric about a vertical line (A H I M O T U V W X Y). MOM qualifies; MAD, TOP (P) and HAM do not.
5	C	In a top/bottom mirror every letter must be symmetric about a horizontal line (B C D E H I K O X). BOOK qualifies; T, M and L do not.
6	B	Side mirror: the flag (nearest the mirror) moves to the far left, the arrow (farthest) to the far right; every shape is flipped left-right and heights stay the same. A keeps the new positions but not the flips; C flips up-down; D is the 180° turn.
7	D	Left and right columns swap, rows stay in place (L and the triangle stay on top and bottom as before), and each shape is flipped left-right. A swaps positions only; B flips up-down; C is the 180° turn.
8	A	Mirror below: top and bottom swap (the triangle moves to the top, the flag to the bottom), left-right positions stay, and every shape is turned upside-down. B swaps positions only; C is the side-mirror image; D is the 180° turn.

CHAPTER 6

Quick Revision

One page for the last day

Mirror position	What swaps	Words	Figures
Right or left	Left-right	Reverse the order, flip each letter	Swap left-right, arrows reverse, slant reverses, up/down same
Above or below	Top-bottom	Same order, each letter upside-down	Swap top-bottom, left-right same

Symmetry	Characters
Vertical axis (unchanged in a side mirror)	A H I M O T U V W X Y; digits 0, 1, 8
Horizontal axis (unchanged in a top/bottom mirror)	B C D E H I K O X; digits 0, 1, 3, 8
Same in side mirror (words)	Palindromes made only of vertical-axis letters, e.g. MOM, TOOT

PYQ RULE: All 8 PYQs: mirror at the right. Method: reverse the order, flip each part, check the part nearest the mirror.

ROUTINE: Exam-day routine: find the mirror position; write the reversed order; flip each asymmetric part; reject unchanged, 180° and up-down options. 30 seconds per question. Next in the syllabus order: Figure Matrix, Paper Folding.